

CHAPTER II

OVERVIEW OF THE SCIENTIFIC METHOD

Here is the abstract of a 2014 article in the journal *Psychological Science*.

Taking notes on laptops rather than in longhand is increasingly common. Many researchers have suggested that laptop note taking is less effective than longhand note taking for learning. Prior studies have primarily focused on students' capacity for multitasking and distraction when using laptops. The present research suggests that even when laptops are used solely to take notes, they may still be impairing learning because their use results in shallower processing. In three studies, we found that students who took notes on laptops performed worse on conceptual questions than students who took notes longhand. We show that whereas taking more notes can be beneficial, laptop note takers' tendency to transcribe lectures verbatim rather than processing information and reframing it in their own words is detrimental to learning. (Mueller & Oppenheimer, 2014, p. 1159)¹

In this abstract, the researcher has identified a research question—about the effect of taking notes on a laptop on learning—and identified why it is worthy of investigation—because the practice is ubiquitous and may be harmful to learning. In this chapter, we give you a broad overview of the various stages of the research process. These include finding a topic of investigation, reviewing the literature, refining your research question and generating a hypothesis, designing and conducting a study, analyzing the data, coming to conclusions, and reporting the results.

Notes

1. Mueller, P. A., & Oppenheimer, D. M. (2014). The pen is mightier than the keyboard: Advantages of longhand over laptop note taking. *Psychological Science*, 25(6), 1159–1168.

7. A Model of Scientific Research in Psychology

Learning Objectives

1. Review a general model of scientific research in psychology.

Figure 2.1 presents a simple model of scientific research in psychology. The researchers formulate a research question, conduct an empirical study designed to answer the question, analyze the resulting data, draw conclusions about the answer to the question, and publishes the results so that they become part of the research literature (i.e., all the published research in that field). Because the research literature is one of the primary sources of new research questions, this process can be thought of as a cycle. New research leads to new questions, which lead to new research, and so on. Figure 2.1 also indicates that research questions can originate outside of this cycle either with informal observations or with practical problems that need to be solved. But even in these cases, the researcher would start by checking the research literature to see if the question had already been answered and to refine it based on what previous research had already found.

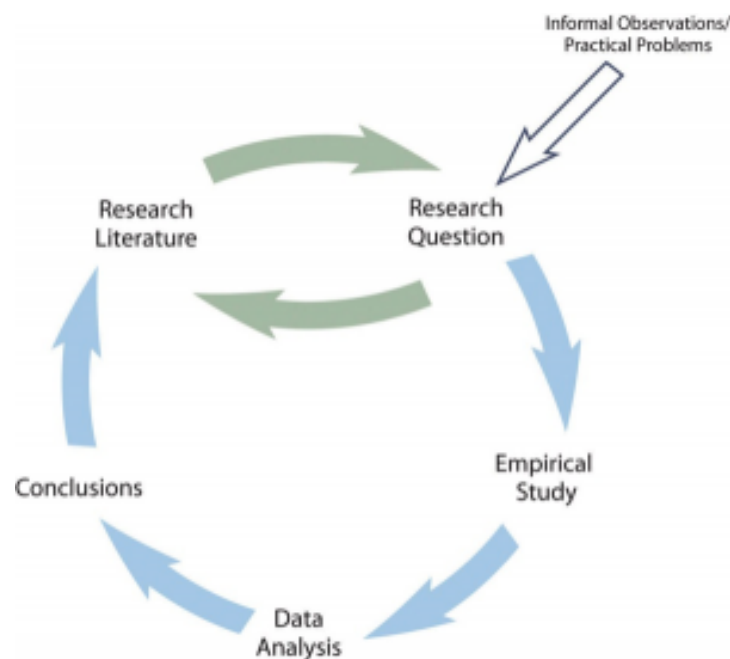


Figure 2.1 A Simple Model of Scientific Research in Psychology

The research by Mehl and his colleagues is described nicely by this model. Their research question—whether women are more talkative than men—was suggested to them both by people’s stereotypes and by claims published in the research literature about the relative talkativeness of women and men. When they checked the research literature, however, they found that this question had not been adequately addressed in scientific studies. They then conducted a careful empirical study, analyzed the results (finding very little difference between women and men), formed their conclusions, and published their work so that it became part of the research literature. The publication of their article is not the end of the story, however, because their work suggests many new questions (about the reliability of the result, about potential cultural differences, etc.) that will likely be taken up by them and by other researchers inspired by their work.



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As another example, consider that as cell phones became more widespread during the 1990s, people began to wonder whether, and to what extent, cell phone use had a negative effect on driving. Many psychologists decided to tackle this question scientifically (e.g., Collet, Guillot, & Petit, 2010)¹. It was clear from previously published research that engaging in a simple verbal task impairs performance on a perceptual or motor task carried out at the same time, but no one had studied the effect specifically of cell phone use on driving. Under carefully controlled conditions, these researchers compared people’s driving performance while using a cell phone with their performance while not using a cell phone, both in the lab and on the road. They found that people’s ability to detect road hazards, reaction time, and maintain control of the vehicle were all impaired by cell phone use. Each new study was published and became part of the growing research literature on this topic. For instance, other research teams subsequently demonstrated that cell phone conversations carry a greater risk than conversations with a passenger who is aware of driving conditions, which often become a point of conversation (e.g., Drews, Pasupathi, & Strayer, 2004)².



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Video Attributions

- “[Understanding driver distraction](#)” by [American Psychological Association](#). Standard YouTube Licence.

Notes

1. Collet, C., Guillot, A., & Petit, C. (2010). Phoning while driving I: A review of epidemiological, psychological, behavioral and physiological studies. *Ergonomics*, 53, 589–601.
2. Drews, F. A., Pasupathi, M., & Strayer, D. L. (2004). Passenger and cell-phone conversations in simulated driving. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 48, 2210–2212.

8. Finding a Research Topic

Learning Objectives

1. Learn some common sources of research ideas.
2. Define the research literature in psychology and give examples of sources that are part of the research literature and sources that are not.
3. Describe and use several methods for finding previous research on a particular research idea or question.

Good research must begin with a good research question. Yet coming up with good research questions is something that novice researchers often find difficult and stressful. One reason is that this is a creative process that can appear mysterious—even magical—with experienced researchers seeming to pull interesting research questions out of thin air. However, psychological research on creativity has shown that it is neither as mysterious nor as magical as it appears. It is largely the product of ordinary thinking strategies and persistence (Weisberg, 1993)¹. This section covers some fairly simple strategies for finding general research ideas, turning those ideas into empirically testable research questions, and finally evaluating those questions in terms of how interesting they are and how feasible they would be to answer.

Finding Inspiration

Research questions often begin as more general research ideas—usually focusing on some behavior or psychological characteristic: talkativeness, learning, depression, bungee jumping, and so on. Before looking at how to turn such ideas into empirically testable research questions, it is worth looking at where such ideas come from in the first place. Three of the most common sources of inspiration are informal observations, practical problems, and previous research.

Informal observations include direct observations of our own and others' behavior as well as secondhand observations from non-scientific sources such as newspapers, books, blogs, and so on. For example, you might notice that you always seem to be in the slowest moving line at the grocery store. Could it be that most people think the same thing? Or you might read in a local newspaper about people donating money and food to a local family whose house has burned down and begin to wonder about who makes such donations and why. Some of the most famous research in psychology has been inspired by informal observations. Stanley Milgram's famous research on obedience to authority, for example, was inspired in part by journalistic reports of the trials of accused Nazi war criminals—many of whom claimed that they

were only obeying orders. This led him to wonder about the extent to which ordinary people will commit immoral acts simply because they are ordered to do so by an authority figure (Milgram, 1963)².

Practical problems can also inspire research ideas, leading directly to applied research in such domains as law, health, education, and sports. Does taking lecture notes by hand improve students' exam performance? How effective is psychotherapy for depression compared to drug therapy? To what extent do cell phones impair people's driving ability? How can we teach children to read more efficiently? What is the best mental preparation for running a marathon?

Probably the most common inspiration for new research ideas, however, is previous research. Recall that science is a kind of large-scale collaboration in which many different researchers read and evaluate each other's work and conduct new studies to build on it. Of course, experienced researchers are familiar with previous research in their area of expertise and probably have a long list of ideas. This suggests that novice researchers can find inspiration by consulting with a more experienced researcher (e.g., students can consult a faculty member). But they can also find inspiration by picking up a copy of almost any professional journal and reading the titles and abstracts. In one typical issue of *Psychological Science*, for example, you can find articles on the perception of shapes, anti-Semitism, police lineups, the meaning of death, second-language learning, people who seek negative emotional experiences, and many other topics. If you can narrow your interests down to a particular topic (e.g., memory) or domain (e.g., health care), you can also look through more specific journals, such as *Memory & Cognition* or *Health Psychology*.



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Reviewing the Research Literature

Once again, one of the most common sources of inspiration is previous research. Therefore, it is important to review the literature early in the research process. The **research literature** in any field is all the published research in that field. Reviewing the research literature means finding, reading, and summarizing the published research relevant to your topic of interest. In addition to helping you discover new research questions, reviewing the literature early in the research process can help you in several other ways.

- It can tell you if a research question has already been answered.
- It can help you evaluate the interestingness of a research question.
- It can give you ideas for how to conduct your own study.
- It can tell you how your study fits into the research literature.



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The research literature in psychology is enormous—including millions of scholarly articles and books dating to the beginning of the field—and it continues to grow. Although its boundaries are somewhat fuzzy, the research literature definitely does not include self-help and other pop psychology books, dictionary and encyclopedia entries, websites, and similar sources that are intended mainly for the general public. These are considered unreliable because they are not reviewed by other researchers and are often based on little more than common sense or personal experience. Wikipedia contains much valuable information, but because its authors are anonymous and may not have any formal training or expertise in that subject area, and its content continually changes it is unsuitable as a basis of sound scientific research. For our purposes, it helps to define the research literature as consisting almost entirely of two types of sources: articles in professional journals, and scholarly books in psychology and related fields.

Professional Journals

Professional journals are periodicals that publish original research articles. There are thousands of professional journals that publish research in psychology and related fields. They are usually published monthly or quarterly in individual issues, each of which contains several articles. The issues are organized into volumes, which usually consist of all the issues for a calendar year. Some journals are published in hard copy only, others in both hard copy and electronic form, and still others in electronic form only.

Most articles in professional journals are one of two basic types: empirical research reports and review articles. **Empirical research reports** describe one or more new empirical studies conducted by the authors. They introduce a research question, explain why it is interesting, review previous research, describe their method and results, and draw their conclusions. **Review articles** summarize previously published research on a topic and usually present new ways to organize or explain the results. When a review article is devoted

primarily to presenting a new theory, it is often referred to as a **theoretical article**. When a review article provides a statistical summary of all of the previous results it is referred to as a **meta-analysis**.



Figure 2.2 Small Sample of the Thousands of Professional Journals That Publish Research in Psychology and Related Fields

Most professional journals in psychology undergo a process of **double-blind peer review**. Researchers who want to publish their work in the journal submit a manuscript to the editor—who is generally an established researcher too—who in turn sends it to two or three experts on the topic. Each reviewer reads the manuscript, writes a critical but constructive review, and sends the review back to the editor along with recommendations about whether the manuscript should be published or not. The editor then decides whether to accept the article for publication, ask the authors to make changes and resubmit it for further consideration, or reject it outright. In any case, the editor forwards the reviewers' written comments to the researchers so that they can revise their manuscript accordingly. This entire process is double-blind, as the reviewers do not know the identity of the researcher(s) and vice versa. Double-blind peer review is helpful because it ensures that the work meets basic standards of the field before it can enter the research literature. However, in order to increase transparency and accountability, some newer open access journals

(e.g., [Frontiers in Psychology](#)) utilize an open peer review process wherein the identities of the reviewers (which remain concealed during the peer review process) are published alongside the journal article.

Scholarly Books

Scholarly books are books written by researchers and practitioners mainly for use by other researchers and practitioners. A **monograph** is written by a single author or a small group of authors and usually, gives a coherent presentation of a topic much like an extended review article. **Edited volumes** have an editor or a small group of editors who recruit many authors to write separate chapters on different aspects of the same topic. Although edited volumes can also give a coherent presentation of the topic, it is not unusual for each chapter to take a different perspective or even for the authors of different chapters to openly disagree with each other. In general, scholarly books undergo a peer review process similar to that used by professional journals.

Literature Search Strategies

Using PsycINFO and Other Databases

The primary method used to search the research literature involves using one or more electronic databases. These include Academic Search Premier, JSTOR, and ProQuest for all academic disciplines, ERIC for education, and PubMed for medicine and related fields. The most important for our purposes, however, is PsycINFO, which is produced by the American Psychological Association (APA). **PsycINFO** is so comprehensive—covering thousands of professional journals and scholarly books going back more than 100 years—that for most purposes its content is synonymous with the research literature in psychology. Like most such databases, PsycINFO is usually available through your university library.

PsycINFO consists of individual records for each article, book chapter, or book in the database. Each record includes basic publication information, an abstract or summary of the work (like the one presented at the start of this chapter), and a list of other works cited by that work. A computer interface allows entering one or more search terms and returns any records that contain those search terms. (These interfaces are provided by different vendors and therefore can look somewhat different depending on the library you use.) Each record also contains lists of keywords that describe the content of the work and also a list of index terms. The index terms are especially helpful because they are standardized. Research on differences between females and males, for example, is always indexed under “Human Sex Differences.” Research on note-taking is always indexed under the term “Learning Strategies.” If you do not know the appropriate index terms, PsycINFO includes a thesaurus that can help you find them.

Given that there are nearly four million records in PsycINFO, you may have to try a variety of search terms in different combinations and at different levels of specificity before you find what you are looking for.

Imagine, for example, that you are interested in the question of whether males and females differ in terms of their ability to recall experiences from when they were very young. If you were to enter the search term “memory,” it would return far too many records to look through individually. This is where the thesaurus helps. Entering “memory” into the thesaurus provides several more specific index terms—one of which is “early memories.” While searching for “early memories” among the index terms still returns too many to look through individually—combining it with “human sex differences” as a second search term returns fewer articles, many of which are highly relevant to the topic.

Depending on the vendor that provides the interface to PsycINFO, you may be able to save, print, or e-mail the relevant PsycINFO records. The records might even contain links to full-text copies of the works themselves. (PsycARTICLES is a database that provides full-text access to articles in all journals published by the APA.) If not, and you want a copy of the work, you will have to find out if your library carries the journal or has the book and the hard copy on the library shelves. Be sure to ask a librarian if you need help.

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Using Other Search Techniques

In addition to entering search terms into PsycINFO and other databases, there are several other techniques you can use to search the research literature. First, if you have one good article or book chapter on your topic—a recent review article is best—you can look through the reference list of that article for other relevant articles, books, and book chapters. In fact, you should do this with any relevant article or book chapter you find. You can also start with a classic article or book chapter on your topic, find its record in PsycINFO (by entering the author’s name or article’s title as a search term), and link from there to a list of other works in PsycINFO that cite that classic article. This works because other researchers working on your topic are likely to be aware of the classic article and cite it in their own work. You can also do a general Internet search using search terms related to your topic or the name of a researcher who conducts research on your topic. This might lead you directly to works that are part of the research literature (e.g., articles in open-access journals or posted on researchers’ own websites). The search engine Google Scholar is especially useful for this purpose. A general Internet search might also lead you to websites that are not part of the research literature but might provide references to works that are. Finally, you can talk to people (e.g., your instructor or other faculty members in psychology) who know something about your topic and can suggest relevant articles and book chapters.





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What to Search For

When you do a literature review, you need to be selective. Not every article, book chapter, and book that relates to your research idea or question will be worth obtaining, reading, and integrating into your review. Instead, you want to focus on sources that help you do four basic things: (a) refine your research question, (b) identify appropriate research methods, (c) place your research in the context of previous research, and (d) write an effective research report. Several basic principles can help you find the most useful sources.

First, it is best to focus on recent research, keeping in mind that what counts as recent depends on the topic. For newer topics that are actively being studied, “recent” might mean published in the past year or two. For older topics that are receiving less attention right now, “recent” might mean within the past 10 years. You will get a feel for what counts as recent for your topic when you start your literature search. A good general rule, however, is to start with sources published in the past five years. The main exception to this rule would be classic articles that turn up in the reference list of nearly every other source. If other researchers think that this work is important, even though it is old, then, by all means, you should include it in your review.

Second, you should look for review articles on your topic because they will provide a useful overview of it—often discussing important definitions, results, theories, trends, and controversies—giving you a good sense of where your own research fits into the literature. You should also look for empirical research reports addressing your question or similar questions, which can give you ideas about how to measure your variables and collect your data. As a general rule, it is good to use methods that others have already used successfully unless you have good reasons not to. Finally, you should look for sources that provide information that can help you argue for the interestingness of your research question. For a study on the effects of cell phone use on driving ability, for example, you might look for information about how widespread cell phone use is, how frequent and costly motor vehicle crashes are, and so on.

How many sources are enough for your literature review? This is a difficult question because it depends on how extensively your topic has been studied and also on your own goals. One study found that across a variety of professional journals in psychology, the average number of sources cited per article was about 50 (Adair & Vohra, 2003)³. This gives a rough idea of what professional researchers consider to be adequate. As a student, you might be assigned a much lower minimum number of references to include, but the principles for selecting the most useful ones remain the same.



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- “[Sample PsycINFO Search on EBSCOhost](#)” by [APA Publishing Training](#). Standard YouTube Licence.
- “[Using Google Scholar \(CLIP\)](#)” by [clipinfolit](#). [CC BY \(Attribution\)](#)

Notes

1. Weisberg, R. W. (1993). *Creativity: Beyond the myth of genius*. New York, NY: Freeman.
2. Milgram, S. (1963). Behavioral study of obedience. *Journal of Abnormal and Social Psychology*, 67, 371–378.
3. Adair, J. G., & Vohra, N. (2003). The explosion of knowledge, references, and citations: Psychology’s unique response to a crisis. *American Psychologist*, 58, 15–23.

9. Generating Good Research Questions

Learning Objectives

1. Describe some techniques for turning research ideas into empirical research questions and use those techniques to generate questions.
2. Explain what makes a research question interesting and evaluate research questions in terms of their interestingness.

Generating Empirically Testable Research Questions

Once you have a research idea, you need to use it to generate one or more empirically testable research questions, that is, questions expressed in terms of a single variable or relationship between variables. One way to do this is to look closely at the discussion section in a recent research article on the topic. This is the last major section of the article, in which the researchers summarize their results, interpret them in the context of past research, and suggest directions for future research. These suggestions often take the form of specific research questions, which you can then try to answer with additional research. This can be a good strategy because it is likely that the suggested questions have already been identified as interesting and important by experienced researchers.

But you may also want to generate your own research questions. How can you do this? First, if you have a particular behavior or psychological characteristic in mind, you can simply conceptualize it as a variable and ask how frequent or intense it is. How many words on average do people speak per day? How accurate are our memories of traumatic events? What percentage of people have sought professional help for depression? If the question has never been studied scientifically—which is something that you will learn when you conduct your literature review—then it might be interesting and worth pursuing.

If scientific research has already answered the question of how frequent or intense the behavior or characteristic is, then you should consider turning it into a question about a relationship between that behavior or characteristic and some other variable. One way to do this is to ask yourself the following series of more general questions and write down all the answers you can think of.

- What are some possible causes of the behavior or characteristic?
- What are some possible effects of the behavior or characteristic?
- What types of people might exhibit more or less of the behavior or characteristic?
- What types of situations might elicit more or less of the behavior or characteristic?

In general, each answer you write down can be conceptualized as a second variable, suggesting a question about a relationship. If you were interested in talkativeness, for example, it might occur to you that a possible cause of this psychological characteristic is family size. Is there a relationship between family size and talkativeness? Or it might occur to you that people seem to be more talkative in same-sex groups than mixed-sex groups. Is there a difference in the average level of talkativeness of people in same-sex groups and people in mixed-sex groups? This approach should allow you to generate many different empirically testable questions about almost any behavior or psychological characteristic.

If through this process you generate a question that has never been studied scientifically—which again is something that you will learn in your literature review—then it might be interesting and worth pursuing. But what if you find that it has been studied scientifically? Although novice researchers often want to give up and move on to a new question at this point, this is not necessarily a good strategy. For one thing, the fact that the question has been studied scientifically and the research published suggests that it is of interest to the scientific community. For another, the question can almost certainly be refined so that its answer will still contribute something new to the research literature. Again, asking yourself a series of more general questions about the relationship is a good strategy.

- Are there other ways to define and measure the variables?
- Are there types of people for whom the relationship might be stronger or weaker?
- Are there situations in which the relationship might be stronger or weaker—including situations with practical importance?

For example, research has shown that women and men speak about the same number of words per day—but this was when talkativeness was measured in terms of the number of words spoken per day among university students in the United States and Mexico. We can still ask whether other ways of measuring talkativeness—perhaps the number of different people spoken to each day—produce the same result. Or we can ask whether studying elderly people or people from other cultures produces the same result. Again, this approach should help you generate many different research questions about almost any relationship.

Evaluating Research Questions

Researchers usually generate many more research questions than they ever attempt to answer. This means they must have some way of evaluating the research questions they generate so that they can choose which ones to pursue. In this section, we consider two criteria for evaluating research questions: the interestingness of the question and the feasibility of answering it.

Interestingness

How often do people tie their shoes? Do people feel pain when you punch them in the jaw? Are women more likely to wear makeup than men? Do people prefer vanilla or chocolate ice cream? Although it would

be a fairly simple matter to design a study and collect data to answer these questions, you probably would not want to because they are not interesting. We are not talking here about whether a research question is interesting to us personally but whether it is interesting to people more generally and, especially, to the scientific community. But what makes a research question interesting in this sense? Here we look at three factors that affect the **interestingness** of a research question: the answer is in doubt, the answer fills a gap in the research literature, and the answer has important practical implications.

First, a research question is interesting to the extent that its answer is in doubt. Obviously, questions that have been answered by scientific research are no longer interesting as the subject of new empirical research. But the fact that a question has not been answered by scientific research does not necessarily make it interesting. There has to be some reasonable chance that the answer to the question will be something that we did not already know. But how can you assess this before actually collecting data? One approach is to try to think of reasons to expect different answers to the question—especially ones that seem to conflict with common sense. If you can think of reasons to expect at least two different answers, then the question might be interesting. If you can think of reasons to expect only one answer, then it probably is not. The question of whether women are more talkative than men is interesting because there are reasons to expect both answers. The existence of the stereotype itself suggests the answer could be yes, but the fact that women's and men's verbal abilities are fairly similar suggests the answer could be no. The question of whether people feel pain when you punch them in the jaw is not interesting because there is absolutely no reason to think that the answer could be anything other than a resounding yes.

A second important factor to consider when deciding if a research question is interesting is whether answering it will fill a gap in the research literature. Again, this means in part that the question has not already been answered by scientific research. But it also means that the question is in some sense a natural one for people who are familiar with the research literature. For example, the question of whether taking lecture notes by hand can help improve students' exam performance would be likely to occur to anyone who was familiar with research on note taking and the ineffectiveness of shallow processing on learning.

A final factor to consider when deciding whether a research question is interesting is whether its answer has important practical implications. Again, the question of whether taking notes by hand improves learning has important implications for education, including classroom policies concerning technology use. The question of whether cell phone use impairs driving is interesting because it is relevant to the personal safety of everyone who travels by car and to the debate over whether cell phone use should be restricted by law.

Feasibility

A second important criterion for evaluating research questions is the **feasibility** of successfully answering them. There are many factors that affect feasibility, including time, money, equipment and materials, technical knowledge and skill, and access to research participants. Clearly, researchers need to take these factors into account so that they do not waste time and effort pursuing research that they cannot complete successfully.

Looking through a sample of professional journals in psychology will reveal many studies that are

complicated and difficult to carry out. These include longitudinal designs in which participants are tracked over many years, neuroimaging studies in which participants' brain activity is measured while they carry out various mental tasks, and complex non-experimental studies involving several variables and complicated statistical analyses. Keep in mind, though, that such research tends to be carried out by teams of highly trained researchers whose work is often supported in part by government and private grants. Also, keep in mind that research does not have to be complicated or difficult to produce interesting and important results. Looking through a sample of professional journals will also reveal studies that are relatively simple and easy to carry out—perhaps involving a convenience sample of university students and a paper-and-pencil task.

A final point here is that it is generally good practice to use methods that have already been used successfully by other researchers. For example, if you want to manipulate people's moods to make some of them happy, it would be a good idea to use one of the many approaches that have been used successfully by other researchers (e.g., paying them a compliment). This is good not only for the sake of feasibility—the approach is “tried and true”—but also because it provides greater continuity with previous research. This makes it easier to compare your results with those of other researchers and to understand the implications of their research for yours, and vice versa.

10. Developing a Hypothesis

Learning Objectives

1. Distinguish between a theory and a hypothesis.
2. Discover how theories are used to generate hypotheses and how the results of studies can be used to further inform theories.
3. Understand the characteristics of a good hypothesis.

Theories and Hypotheses

Before describing how to develop a hypothesis, it is important to distinguish between a theory and a hypothesis. A **theory** is a coherent explanation or interpretation of one or more phenomena. Although theories can take a variety of forms, one thing they have in common is that they go beyond the phenomena they explain by including variables, structures, processes, functions, or organizing principles that have not been observed directly. Consider, for example, Zajonc's theory of social facilitation and social inhibition (1965)¹. He proposed that being watched by others while performing a task creates a general state of physiological arousal, which increases the likelihood of the dominant (most likely) response. So for highly practiced tasks, being watched increases the tendency to make correct responses, but for relatively unpracticed tasks, being watched increases the tendency to make incorrect responses. Notice that this theory—which has come to be called drive theory—provides an explanation of both social facilitation and social inhibition that goes beyond the phenomena themselves by including concepts such as “arousal” and “dominant response,” along with processes such as the effect of arousal on the dominant response.

Outside of science, referring to an idea as a theory often implies that it is untested—perhaps no more than a wild guess. In science, however, the term theory has no such implication. A theory is simply an explanation or interpretation of a set of phenomena. It can be untested, but it can also be extensively tested, well supported, and accepted as an accurate description of the world by the scientific community. The theory of evolution by natural selection, for example, is a theory because it is an explanation of the diversity of life on earth—not because it is untested or unsupported by scientific research. On the contrary, the evidence for this theory is overwhelmingly positive and nearly all scientists accept its basic assumptions as accurate. Similarly, the “germ theory” of disease is a theory because it is an explanation of the origin of various diseases, not because there is any doubt that many diseases are caused by microorganisms that infect the body.

A **hypothesis**, on the other hand, is a specific prediction about a new phenomenon that should be observed if a particular theory is accurate. It is an explanation that relies on just a few key concepts. Hypotheses are

often specific predictions about what will happen in a particular study. They are developed by considering existing evidence and using reasoning to infer what will happen in the specific context of interest. Hypotheses are often but not always derived from theories. So a hypothesis is often a prediction based on a theory but some hypotheses are a-theoretical and only after a set of observations have been made, is a theory developed. This is because theories are broad in nature and they explain larger bodies of data. So if our research question is really original then we may need to collect some data and make some observations before we can develop a broader theory.

Theories and hypotheses always have this *if-then* relationship. “If drive theory is correct, *then* cockroaches should run through a straight runway faster, and a branching runway more slowly, when other cockroaches are present.” Although hypotheses are usually expressed as statements, they can always be rephrased as questions. “Do cockroaches run through a straight runway faster when other cockroaches are present?” Thus deriving hypotheses from theories is an excellent way of generating interesting research questions.

But how do researchers derive hypotheses from theories? One way is to generate a research question using the techniques discussed in this chapter and then ask whether any theory implies an answer to that question. For example, you might wonder whether expressive writing about positive experiences improves health as much as expressive writing about traumatic experiences. Although this question is an interesting one on its own, you might then ask whether the habituation theory—the idea that expressive writing causes people to habituate to negative thoughts and feelings—implies an answer. In this case, it seems clear that if the habituation theory is correct, then expressive writing about positive experiences should not be effective because it would not cause people to habituate to negative thoughts and feelings. A second way to derive hypotheses from theories is to focus on some component of the theory that has not yet been directly observed. For example, a researcher could focus on the process of habituation—perhaps hypothesizing that people should show fewer signs of emotional distress with each new writing session.

Among the very best hypotheses are those that distinguish between competing theories. For example, Norbert Schwarz and his colleagues considered two theories of how people make judgments about themselves, such as how assertive they are (Schwarz et al., 1991)². Both theories held that such judgments are based on relevant examples that people bring to mind. However, one theory was that people base their judgments on the *number* of examples they bring to mind and the other was that people base their judgments on how *easily* they bring those examples to mind. To test these theories, the researchers asked people to recall either six times when they were assertive (which is easy for most people) or 12 times (which is difficult for most people). Then they asked them to judge their own assertiveness. Note that the number-of-examples theory implies that people who recalled 12 examples should judge themselves to be more assertive because they recalled more examples, but the ease-of-examples theory implies that participants who recalled six examples should judge themselves as more assertive because recalling the examples was easier. Thus the two theories made opposite predictions so that only one of the predictions could be confirmed. The surprising result was that participants who recalled fewer examples judged themselves to be more assertive—providing particularly convincing evidence in favor of the ease-of-retrieval theory over the number-of-examples theory.

Theory Testing

The primary way that scientific researchers use theories is sometimes called the **hypothetico-deductive method** (although this term is much more likely to be used by philosophers of science than by scientists themselves). Researchers begin with a set of phenomena and either construct a theory to explain or interpret them or choose an existing theory to work with. They then make a prediction about some new phenomenon that should be observed if the theory is correct. Again, this prediction is called a hypothesis. The researchers then conduct an empirical study to test the hypothesis. Finally, they reevaluate the theory in light of the new results and revise it if necessary. This process is usually conceptualized as a cycle because the researchers can then derive a new hypothesis from the revised theory, conduct a new empirical study to test the hypothesis, and so on. As Figure 2.3 shows, this approach meshes nicely with the model of scientific research in psychology presented earlier in the textbook—creating a more detailed model of “theoretically motivated” or “theory-driven” research.

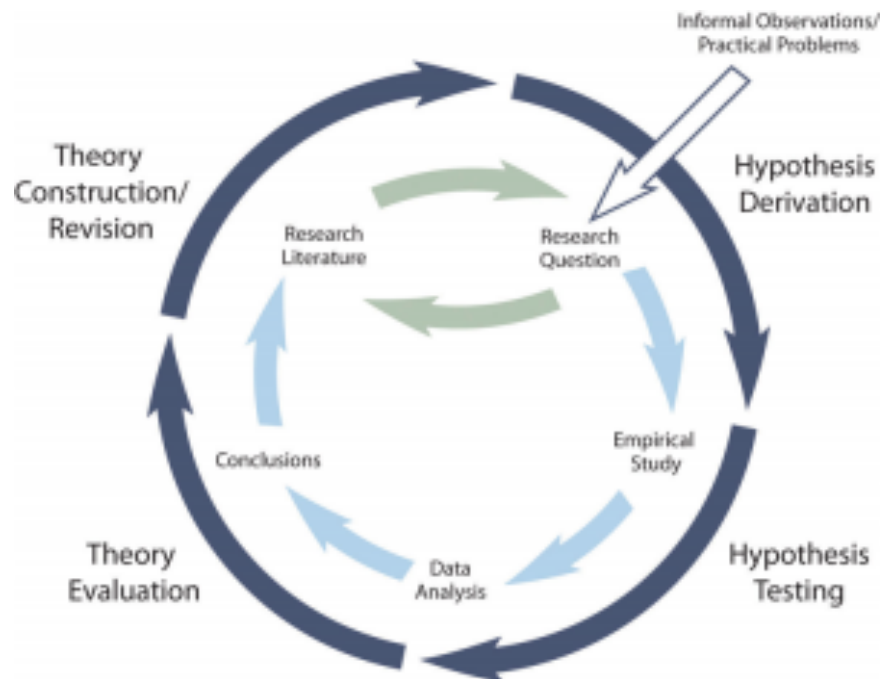


Figure 2.3 Hypothetico-Deductive Method Combined With the General Model of Scientific Research in Psychology Together they form a model of theoretically motivated research.

As an example, let us consider Zajonc’s research on social facilitation and inhibition. He started with a somewhat contradictory pattern of results from the research literature. He then constructed his drive theory, according to which being watched by others while performing a task causes physiological arousal, which increases an organism’s tendency to make the dominant response. This theory predicts social facilitation for well-learned tasks and social inhibition for poorly learned tasks. He now had a theory that organized previous results in a meaningful way—but he still needed to test it. He hypothesized that if

his theory was correct, he should observe that the presence of others improves performance in a simple laboratory task but inhibits performance in a difficult version of the very same laboratory task. To test this hypothesis, one of the studies he conducted used cockroaches as subjects (Zajonc, Heingartner, & Herman, 1969)³. The cockroaches ran either down a straight runway (an easy task for a cockroach) or through a cross-shaped maze (a difficult task for a cockroach) to escape into a dark chamber when a light was shined on them. They did this either while alone or in the presence of other cockroaches in clear plastic “audience boxes.” Zajonc found that cockroaches in the straight runway reached their goal more quickly in the presence of other cockroaches, but cockroaches in the cross-shaped maze reached their goal more slowly when they were in the presence of other cockroaches. Thus he confirmed his hypothesis and provided support for his drive theory. (Zajonc also showed that drive theory existed in humans [Zajonc & Sales, 1966]⁴ in many other studies afterward).

Incorporating Theory into Your Research

When you write your research report or plan your presentation, be aware that there are two basic ways that researchers usually include theory. The first is to raise a research question, answer that question by conducting a new study, and then offer one or more theories (usually more) to explain or interpret the results. This format works well for applied research questions and for research questions that existing theories do not address. The second way is to describe one or more existing theories, derive a hypothesis from one of those theories, test the hypothesis in a new study, and finally reevaluate the theory. This format works well when there is an existing theory that addresses the research question—especially if the resulting hypothesis is surprising or conflicts with a hypothesis derived from a different theory.

To use theories in your research will not only give you guidance in coming up with experiment ideas and possible projects, but it lends legitimacy to your work. Psychologists have been interested in a variety of human behaviors and have developed many theories along the way. Using established theories will help you break new ground as a researcher, not limit you from developing your own ideas.

Characteristics of a Good Hypothesis

There are three general characteristics of a good hypothesis. First, a good hypothesis must be **testable and falsifiable**. We must be able to test the hypothesis using the methods of science and if you’ll recall Popper’s falsifiability criterion, it must be possible to gather evidence that will disconfirm the hypothesis if it is indeed false. Second, a good hypothesis must be **logical**. As described above, hypotheses are more than just a random guess. Hypotheses should be informed by previous theories or observations and logical reasoning. Typically, we begin with a broad and general theory and use *deductive reasoning* to generate a more specific hypothesis to test based on that theory. Occasionally, however, when there is no theory to inform our hypothesis, we use *inductive reasoning* which involves using specific observations or research findings to form a more general hypothesis. Finally, the hypothesis should be **positive**. That is, the hypothesis should make a positive statement about the existence of a relationship or effect, rather than

a statement that a relationship or effect does not exist. As scientists, we don't set out to show that relationships do not exist or that effects do not occur so our hypotheses should not be worded in a way to suggest that an effect or relationship does not exist. The nature of science is to assume that something does not exist and then seek to find evidence to prove this wrong, to show that it really does exist. That may seem backward to you but that is the nature of the scientific method. The underlying reason for this is beyond the scope of this chapter but it has to do with statistical theory.

Notes

1. Zajonc, R. B. (1965). Social facilitation. *Science*, 149, 269–274
2. Schwarz, N., Bless, H., Strack, F., Klumpp, G., Rittenauer-Schatka, H., & Simons, A. (1991). Ease of retrieval as information: Another look at the availability heuristic. *Journal of Personality and Social Psychology*, 61, 195–202.
3. Zajonc, R. B., Heingartner, A., & Herman, E. M. (1969). Social enhancement and impairment of performance in the cockroach. *Journal of Personality and Social Psychology*, 13, 83–92.
4. Zajonc, R.B. & Sales, S.M. (1966). Social facilitation of dominant and subordinate responses. *Journal of Experimental Social Psychology*, 2, 160–168.

II. Designing a Research Study

Learning Objectives

1. Define the concept of a variable, distinguish quantitative from categorical variables, and give examples of variables that might be of interest to psychologists.
2. Explain the difference between a population and a sample.
3. Distinguish between experimental and non-experimental research.
4. Distinguish between lab studies, field studies, and field experiments.

Identifying and Defining the Variables and Population

Variables and Operational Definitions

Part of generating a hypothesis involves identifying the variables that you want to study and operationally defining those variables so that they can be measured. Research questions in psychology are about variables. A **variable** is a quantity or quality that varies across people or situations. For example, the height of the students enrolled in a university course is a variable because it varies from student to student. The chosen major of the students is also a variable as long as not everyone in the class has declared the same major. Almost everything in our world varies and as such thinking of examples of constants (things that don't vary) is far more difficult. A rare example of a constant is the speed of light. Variables can be either quantitative or categorical. A **quantitative variable** is a quantity, such as height, that is typically measured by assigning a number to each individual. Other examples of quantitative variables include people's level of talkativeness, how depressed they are, and the number of siblings they have. A **categorical variable** is a quality, such as chosen major, and is typically measured by assigning a category label to each individual (e.g., Psychology, English, Nursing, etc.). Other examples include people's nationality, their occupation, and whether they are receiving psychotherapy.

After the researcher generates their hypothesis and selects the variables they want to manipulate and measure, the researcher needs to find ways to actually measure the variables of interest. This requires an **operational definition**—a definition of the variable in terms of precisely how it is to be measured. Most variables that researchers are interested in studying cannot be directly observed or measured and this poses a problem because empiricism (observation) is at the heart of the scientific method. Operationally defining a variable involves taking an abstract construct like depression that cannot be directly observed and transforming it into something that can be directly observed and measured. Most variables can be

operationally defined in many different ways. For example, depression can be operationally defined as people's scores on a paper-and-pencil depression scale such as the Beck Depression Inventory, the number of depressive symptoms they are experiencing, or whether they have been diagnosed with major depressive disorder. Researchers are wise to choose an operational definition that has been used extensively in the research literature.

Sampling and Measurement

In addition to identifying which variables to manipulate and measure, and operationally defining those variables, researchers need to identify the population of interest. Researchers in psychology are usually interested in drawing conclusions about some very large group of people. This is called the **population**. It could be all American teenagers, children with autism, professional athletes, or even just human beings—depending on the interests and goals of the researcher. But they usually study only a small subset or **sample** of the population. For example, a researcher might measure the talkativeness of a few hundred university students with the intention of drawing conclusions about the talkativeness of men and women in general. It is important, therefore, for researchers to use a representative sample—one that is similar to the population in important respects.

One method of obtaining a sample is **simple random sampling**, in which every member of the population has an equal chance of being selected for the sample. For example, a pollster could start with a list of all the registered voters in a city (the population), randomly select 100 of them from the list (the sample), and ask those 100 whom they intend to vote for. Unfortunately, random sampling is difficult or impossible in most psychological research because the populations are less clearly defined than the registered voters in a city. How could a researcher give all American teenagers or all children with autism an equal chance of being selected for a sample? The most common alternative to random sampling is **convenience sampling**, in which the sample consists of individuals who happen to be nearby and willing to participate (such as introductory psychology students). Of course, the obvious problem with convenience sampling is that the sample might not be representative of the population and therefore it may be less appropriate to generalize the results from the sample to that population.

Experimental vs. Non-Experimental Research

The next step a researcher must take is to decide which type of approach they will use to collect the data. As you will learn in your research methods course there are many different approaches to research that can be divided in many different ways. One of the most fundamental distinctions is between experimental and non-experimental research.

Experimental Research

Researchers who want to test hypotheses about causal relationships between variables (i.e., their goal is to explain) need to use an experimental method. This is because the experimental method is the only method that allows us to determine causal relationships. Using the experimental approach, researchers first manipulate one or more variables while attempting to control extraneous variables, and then they measure how the manipulated variables affect participants' responses.

The terms independent variable and dependent variable are used in the context of experimental research. The **independent variable** is the variable the experimenter manipulates (it is the presumed cause) and the **dependent variable** is the variable the experimenter measures (it is the presumed effect).

Extraneous variables are any variable other than the dependent variable. **Confounds** are a specific type of extraneous variable that systematically varies along with the variables under investigation and therefore provides an alternative explanation for the results. When researchers design an experiment they need to ensure that they control for confounds; they need to ensure that extraneous variables don't become confounding variables because in order to make a causal conclusion they need to make sure alternative explanations for the results have been ruled out.

As an example, if we manipulate the lighting in the room and examine the effects of that manipulation on workers' productivity, then the lighting conditions (bright lights vs. dim lights) would be considered the independent variable and the workers' productivity would be considered the dependent variable. If the bright lights are noisy then that noise would be a confound since the noise would be present whenever the lights are bright and the noise would be absent when the lights are dim. If noise is varying systematically with light then we wouldn't know if a difference in worker productivity across the two lighting conditions is due to noise or light. So confounds are bad, they disrupt our ability to make causal conclusions about the nature of the relationship between variables. However, if there is noise in the room both when the lights are on and when the lights are off then noise is merely an extraneous variable (it is a variable other than the independent or dependent variable) and we don't worry much about extraneous variables. This is because unless a variable varies systematically with the manipulated independent variable it cannot be a competing explanation for the results.

Non-Experimental Research

Researchers who are simply interested in describing characteristics of people, describing relationships between variables, and using those relationships to make predictions can use non-experimental research. Using the non-experimental approach, the researcher simply measures variables as they naturally occur, but they do not manipulate them. For instance, if I just measured the number of traffic fatalities in America last year that involved the use of a cell phone but I did not actually manipulate cell phone use then this would be categorized as non-experimental research. Alternatively, if I stood at a busy intersection and recorded drivers' genders and whether or not they were using a cell phone when they passed through the intersection to see whether men or women are more likely to use a cell phone when driving, then this would be non-

experimental research. It is important to point out that non-experimental does not mean nonscientific. Non-experimental research is scientific in nature. It can be used to fulfill two of the three goals of science (to describe and to predict). However, unlike with experimental research, we cannot make causal conclusions using this method; we cannot say that one variable causes another variable using this method.

Laboratory vs. Field Research

The next major distinction between research methods is between laboratory and field studies. A **laboratory study** is a study that is conducted in the laboratory environment. In contrast, a **field study** is a study that is conducted in the real-world, in a natural environment.

Laboratory experiments typically have high **internal validity**. Internal validity refers to the degree to which we can confidently infer a causal relationship between variables. When we conduct an experimental study in a laboratory environment we have very high internal validity because we manipulate one variable while controlling all other outside extraneous variables. When we manipulate an independent variable and observe an effect on a dependent variable and we control for everything else so that the only difference between our experimental groups or conditions is the one manipulated variable then we can be quite confident that it is the independent variable that is causing the change in the dependent variable. In contrast, because field studies are conducted in the real-world, the experimenter typically has less control over the environment and potential extraneous variables, and this decreases internal validity, making it less appropriate to arrive at causal conclusions.

But there is typically a trade-off between internal and external validity. **External validity** simply refers to the degree to which we can generalize the findings to other circumstances or settings, like the real-world environment. When internal validity is high, external validity tends to be low; and when internal validity is low, external validity tends to be high. So laboratory studies are typically low in external validity, while field studies are typically high in external validity. Since field studies are conducted in the real-world environment it is far more appropriate to generalize the findings to that real-world environment than when the research is conducted in the more artificial sterile laboratory.

Finally, there are field studies which are non-experimental in nature because nothing is manipulated. But there are also **field experiments** where an independent variable is manipulated in a natural setting and extraneous variables are controlled. Depending on their overall quality and the level of control of extraneous variables, such field experiments can have high external and high internal validity.

12. Analyzing the Data

Learning Objectives

1. Distinguish between descriptive and inferential statistics
2. Identify the different kinds of descriptive statistics researchers use to summarize their data
3. Describe the purpose of inferential statistics.
4. Distinguish between Type I and Type II errors.

Once the study is complete and the observations have been made and recorded the researchers need to analyze the data and draw their conclusions. Typically, data are analyzed using both descriptive and inferential statistics. Descriptive statistics are used to summarize the data and inferential statistics are used to generalize the results from the sample to the population. In turn, inferential statistics are used to make conclusions about whether or not a theory has been supported, refuted, or requires modification.

Descriptive Statistics

Descriptive statistics are used to organize or summarize a set of data. Examples include percentages, measures of central tendency (mean, median, mode), measures of dispersion (range, standard deviation, variance), and correlation coefficients.

Measures of central tendency are used to describe the typical, average and center of a distribution of scores. The **mode** is the most frequently occurring score in a distribution. The **median** is the midpoint of a distribution of scores. The **mean** is the average of a distribution of scores.

Measures of dispersion are also considered descriptive statistics. They are used to describe the degree of spread in a set of scores. So are all of the scores similar and clustered around the mean or is there a lot of variability in the scores? The **range** is a measure of dispersion that measures the distance between the highest and lowest scores in a distribution. The **standard deviation** is a more sophisticated measure of dispersion that measures the average distance of scores from the mean. The **variance** is just the standard deviation squared. So it also measures the distance of scores from the mean but in a different unit of measure.

Typically means and standard deviations are computed for experimental research studies in which an independent variable was manipulated to produce two or more groups and a dependent variable was

measured quantitatively. The means from each experimental group or condition are calculated separately and are compared to see if they differ.

For non-experimental research, simple percentages may be computed to describe the percentage of people who engaged in some behavior or held some belief. But more commonly non-experimental research involves computing the correlation between two variables. A **correlation coefficient** describes the strength and direction of the relationship between two variables. The values of a correlation coefficient can range from -1.00 (the strongest possible negative relationship) to +1.00 (the strongest possible positive relationship). A value of 0 means there is no relationship between the two variables. Positive correlation coefficients indicate that as the values of one variable increase, so do the values of the other variable. A good example of a positive correlation is the correlation between height and weight, because as height increases weight also tends to increase. Negative correlation coefficients indicate that as the value of one variable increase, the values of the other variable decrease. An example of a negative correlation is the correlation between stressful life events and happiness; because as stress increases, happiness is likely to decrease.

Inferential Statistics

As you learned in the section of this chapter on sampling, typically researchers sample from a population but ultimately they want to be able to generalize their results from the sample to a broader population. Researchers typically want to infer what the population is like based on the sample they studied. Inferential statistics are used for that purpose. **Inferential statistics** allow researchers to draw conclusions about a population based on data from a sample. Inferential statistics are crucial because the effects (i.e., the differences in the means or the correlation coefficient) that researchers find in a study may be due simply to random chance variability or they may be due to a real effect (i.e., they may reflect a real relationship between variables or a real effect of an independent variable on a dependent variable).

Researchers use inferential statistics to determine whether their effects are statistically significant. A **statistically significant** effect is one that is unlikely due to random chance and therefore likely represents a real effect in the population. More specifically results that have less than a 5% chance of being due to random error are typically considered statistically significant. When an effect is statistically significant it is appropriate to generalize the results from the sample to the population. In contrast, if inferential statistics reveal that there is more than a 5% chance that an effect could be due to chance error alone then the researcher must conclude that their result is not statistically significant.

It is important to keep in mind that statistics are probabilistic in nature. They allow researchers to determine whether the chances are low that their results are due to random error, but they don't provide any absolute certainty. Hopefully, when we conclude that an effect is statistically significant it is a real effect that we would find if we tested the entire population. And hopefully when we conclude that an effect is not statistically significant there really is no effect and if we tested the entire population we would find no effect. And that 5% threshold is set at 5% to ensure that there is a high probability that we make a correct decision and that our determination of statistical significance is an accurate reflection of reality.

But mistakes can always be made. Specifically, two kinds of mistakes can be made. First, researchers can

make a **Type I error**, which is a false positive. It is when a researcher concludes that their results are statistically significant (so they say there is an effect in the population) when in reality there is no real effect in the population and the results are just due to chance (they are a fluke). When the threshold is set to 5%, which is the convention, then the researcher has a 5% chance or less of making a Type I error. You might wonder why researchers don't set it even lower to reduce the chances of making a Type I error. The reason is when the chances of making a Type I error are reduced, the chances of making a Type II error are increased. A **Type II error** is a missed opportunity. It is when a researcher concludes that their results are not statistically significant when in reality there is a real effect in the population and they just missed detecting it. Once again, these Type II errors are more likely to occur when the threshold is set too low (e.g., set at 1% instead of 5%) and/or when the sample was too small.

13. Drawing Conclusions and Reporting the Results

Learning Objectives

1. Identify the conclusions researchers can make based on the outcome of their studies.
2. Describe why scientists avoid the term “scientific proof.”
3. Explain the different ways that scientists share their findings.

Drawing Conclusions

Since statistics are probabilistic in nature and findings can reflect type I or type II errors, we cannot use the results of a single study to conclude with certainty that a theory is true. Rather theories are supported, refuted, or modified based on the results of research.

If the results are statistically significant and consistent with the hypothesis and the theory that was used to generate the hypothesis, then researchers can conclude that the theory is supported. Not only did the theory make an accurate prediction, but there is now a new phenomenon that the theory accounts for. If a hypothesis is disconfirmed in a systematic empirical study, then the theory has been weakened. It made an inaccurate prediction, and there is now a new phenomenon that it does not account for.

Although this seems straightforward, there are some complications. First, confirming a hypothesis can strengthen a theory but it can never prove a theory. In fact, scientists tend to avoid the word “prove” when talking and writing about theories. One reason for this avoidance is that the result may reflect a type I error. Another reason for this avoidance is that there may be other plausible theories that imply the same hypothesis, which means that confirming the hypothesis strengthens all those theories equally. A third reason is that it is always possible that another test of the hypothesis or a test of a new hypothesis derived from the theory will be disconfirmed. This difficulty is a version of the famous philosophical “problem of induction.” One cannot definitively prove a general principle (e.g., “All swans are white.”) just by observing confirming cases (e.g., white swans)—no matter how many. It is always possible that a disconfirming case (e.g., a black swan) will eventually come along. For these reasons, scientists tend to think of theories—even highly successful ones—as subject to revision based on new and unexpected observations.

A second complication has to do with what it means when a hypothesis is disconfirmed. According to the strictest version of the hypothetico-deductive method, disconfirming a hypothesis disproves the theory it was derived from. In formal logic, the premises “if A then B” and “not B” necessarily lead to the conclusion

“not A.” If A is the theory and B is the hypothesis (“if A then B”), then disconfirming the hypothesis (“not B”) must mean that the theory is incorrect (“not A”). In practice, however, scientists do not give up on their theories so easily. One reason is that one disconfirmed hypothesis could be a missed opportunity (the result of a type II error) or it could be the result of a faulty research design. Perhaps the researcher did not successfully manipulate the independent variable or measure the dependent variable.

A disconfirmed hypothesis could also mean that some unstated but relatively minor assumption of the theory was not met. For example, if Zajonc had failed to find social facilitation in cockroaches, he could have concluded that drive theory is still correct but it applies only to animals with sufficiently complex nervous systems. That is, the evidence from a study can be used to modify a theory. This practice does not mean that researchers are free to ignore disconfirmations of their theories. If they cannot improve their research designs or modify their theories to account for repeated disconfirmations, then they eventually must abandon their theories and replace them with ones that are more successful.

The bottom line here is that because statistics are probabilistic in nature and because all research studies have flaws there is no such thing as scientific proof, there is only scientific evidence.

Reporting the Results

The final step in the research process involves reporting the results. As described in the section on Reviewing the Research Literature in this chapter, results are typically reported in peer-reviewed journal articles and at conferences.

The most prestigious way to report one's findings is by writing a manuscript and having it published in a peer-reviewed scientific journal. Manuscripts published in psychology journals typically must adhere to the writing style of the American Psychological Association (APA style). You will likely be learning the major elements of this writing style in this course.

Another way to report findings is by writing a book chapter that is published in an edited book. Preferably the editor of the book puts the chapter through peer review but this is not always the case and some scientists are invited by editors to write book chapters.

A fun way to disseminate findings is to give a presentation at a conference. This can either be done as an oral presentation or a poster presentation. Oral presentations involve getting up in front of an audience of fellow scientists and giving a talk that might last anywhere from 10 minutes to 1 hour (depending on the conference) and then fielding questions from the audience. Alternatively, poster presentations involve summarizing the study on a large poster that provides a brief overview of the purpose, methods, results, and discussion. The presenter stands by their poster for an hour or two and discusses it with people who pass by. Presenting one's work at a conference is a great way to get feedback from one's peers before attempting to undergo the more rigorous peer-review process involved in publishing a journal article.

14. Key Takeaways and Exercise

Key Takeaways

- Research in psychology can be described by a simple cyclical model. A research question based on the research literature leads to an empirical study, the results of which are published and become part of the research literature.
- The research literature in psychology is all the published research in psychology, consisting primarily of articles in professional journals and scholarly books.
- Early in the research process, it is important to conduct a review of the research literature on your topic to refine your research question, identify appropriate research methods, place your question in the context of other research, and prepare to write an effective research report.
- There are several strategies for finding previous research on your topic. Among the best is using PsycINFO, a computer database that catalogs millions of articles, books, and book chapters in psychology and related fields.
- Research questions expressed in terms of variables and relationships between variables can be suggested by other researchers or generated by asking a series of more general questions about the behavior or psychological characteristic of interest.
- It is important to evaluate how interesting a research question is before designing a study and collecting data to answer it. Factors that affect interestingness are the extent to which the answer is in doubt, whether it fills a gap in the research literature, and whether it has important practical implications.
- It is also important to evaluate how feasible a research question will be to answer. Factors that affect feasibility include time, money, technical knowledge and skill, and access to special equipment and research participants.
- A theory is broad in nature and explains larger bodies of data. A hypothesis is more specific and makes a prediction about the outcome of a particular study.
- Working with theories is not “icing on the cake.” It is a basic ingredient of psychological research.
- Like other scientists, psychologists use the hypothetico-deductive method. They construct theories to explain or interpret phenomena (or work with existing theories), derive hypotheses from their theories, test the hypotheses, and then reevaluate the theories in light of the new results.
- Variables vary across people or situations and may be quantitative (e.g., age) or categorical (e.g., course subject).
- A sample is a small subset of a larger population that is selected to participate in the research study. There are many different ways of sampling participants including convenience sampling and simple random sampling.
- Experimental research involves manipulating an independent variable to observe the effects on a measured dependent variable while non-experimental research involves measuring variables as they naturally occur (i.e., without manipulating anything).
- Research can be conducted in the field or the lab. Laboratory experiments tend to have high internal validity (allowing us to make strong causal conclusions), while field studies often have more external validity (allowing us to generalize to the real world).
- The mean, median, and mode are measures of central tendency used to describe the typical, average, or

center scores in a distribution. The range, standard deviation, and variance are measures of how dispersed or spread apart the scores are. Measures of central tendency and dispersion are important descriptive statistics.

- Inferential statistics allow researchers to determine whether their findings are statistically significant, that is, whether they are unlikely to be due to chance alone and therefore are likely to represent a real effect in the population.
- Since statistics are probabilistic in nature we never know if our conclusions are correct. We can make type I errors (concluding an effect is real when it is not) or type II errors (concluding there is no effect when there actually is a real effect in the population).
- Theories can be supported by not proved. Similarly, disconfirming a hypothesis does not necessarily mean that theory has been disproved.
- The final step of the research process involves reporting results at scientific conferences, in journal articles, and/or in books.

Exercises

- Practice: Find a description of an empirical study in a professional journal or in one of the scientific psychology blogs. Then write a brief description of the research in terms of the cyclical model presented here. One or two sentences for each part of the cycle should suffice.
- Watch the following TED Ed video, in which David H. Schwartz provides an introduction to two types of empirical studies along with some methods that scientists use to increase the reliability of their results:



One or more interactive elements has been excluded from this version of the text. You can view them online here: <https://kpu.pressbooks.pub/psychmethods4e/?p=271#oembed-1>

- Reading in print? Go to <https://youtu.be/GUPd2HJHUt8> or scan this QR code with your phone:



- Practice: Use the techniques discussed in this section to find 10 journal articles and book chapters on one of the following research ideas: memory for smells, aggressive driving, the causes of narcissistic personality disorder, the functions of the intraparietal sulcus, or prejudice against the physically handicapped.
- Watch the following video clip produced by UBCiSchool about how to read an academic paper (without

losing your mind):



One or more interactive elements has been excluded from this version of the text. You can view them online here: <https://kpu.pressbooks.pub/psychmethods4e/?p=271#oembed-2>

- Reading in print? Go to https://youtu.be/SKxm2HF_-k0 or scan this QR code with your phone:



- Practice: Generate three research ideas based on each of the following: informal observations, practical problems, and topics discussed in recent issues of professional journals.
- Practice: Generate an empirical research question about each of the following behaviors or psychological characteristics: long-distance running, getting tattooed, social anxiety, bullying, and memory for early childhood events.
- Practice: Evaluate each of the research questions you generated in Exercise 2 in terms of its interestingness based on the criteria discussed in this section.
- Practice: Find an issue of a journal that publishes short empirical research reports (e.g., *Psychological Science*, *Psychonomic Bulletin and Review*, *Personality and Social Psychology Bulletin*). Pick three studies, and rate each one in terms of how feasible it would be for you to replicate it with the resources available to you right now. Use the following rating scale: (1) You could replicate it essentially as reported. (2) You could replicate it with some simplifications. (3) You could not replicate it. Explain each rating.
- Practice: Find a recent empirical research report in a professional journal. Read the introduction and highlight in different colors descriptions of theories and hypotheses.
- Practice: Using the research article you found in a professional journal identify whether the study was experimental or non-experimental. If it was experimental identify the independent and dependent variables.
- Practice: Using the research article you found in a professional journal identify which descriptive statistics were reported.
- Practice: Describe why theories can be supported but not proved.

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